Fiscal Year:	FY 2023	**Task Last Updated:**	FY 06/11/2023
PI Name:	Li, Chen Ph.D.		
Project Title:	Impacts of Gravity on Dropwise Condensation-Enhanced Heat Pipes		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	29208	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	EPSCoR-Physical Sciences NNH20ZHA001C
Start Date:	06/01/2020	**End Date:**	05/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA EPSCoR
Contact Monitor:	Compton, Jeppie	**Contact Phone:**	
Contact Email:	jeppie.r.compton@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2022 (original end date was 5/31/2021) per NSSC information (Ed., 1/20/22)		
Key Personnel Changes/Previous PI:	Dr. Cassandra J. Runyon serves as the South Carolina NASA EPSCoR Director; she is the administrative Principal Investigator on this proposal and will oversee the overall progress of the work and report to NASA.		
COI Name (Institution):	Chao, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0153		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Established Program to Stimulate Competitive Research (EPSCoR) Project Primarily consisting of evaporation and condensation sections, the heat pipe is a light, highly efficient, and durable two-phase device that transports heat over a distance with a small temperature drop and a type of primary devices in thermal control systems of satellites, space shuttles, and crewed space stations. Ultrahigh-performance, long, and light-weight heat pipes are critical to future space missions (e.g., crewed missions to Mars) and hence, highly desired. Numerous efforts have been made to improve heat pipe performance by engineering the evaporation section and are nearly approaching the physical limits. However, the heat pipe condensation section that results in an order of magnitude higher thermal resistance than the evaporation one has not been well developed. It is promising if a performance leap of heat pipe technologies can be realized by enhancing the condensation section. Heat transfer rate of dropwise condensation (DWC) is 10 times higher than film wise condensation adopted in existing heat pipe technologies. Most recently, using super-durable Ni-graphene coatings in a terrestrial environment, the Science Principal Investigator's team has successfully demonstrated 8 times higher effective thermal conductivity by promoting DWC inside heat pipes. However, the enhancement was highly sensitive to orientations. In most space environments, due to the reduced gravity, droplets would attach on the wall and DWC could totally fail without quickly removing droplets. The objective of this proposed project is to explore a feasibility in implementing DWC for space missions. Conducting experiments in a drop tower would be the first step to verify the feasibility of wick designs in rapidly removing droplets inside heat pipes. In this proposal, condensation sections of heat pipes will be coated with the demonstrated Ni-graphene coatings. Moreover, various wicking structures will be designed to eventually realize rapid droplet removal in a microgravity environment. The gap between the Ni-graphene coated wall and wicking structures would play a critical role in maximizing DWC and hence, heat pipe performances. Heat pipes with various wicking structure designs and gaps will be evaluated before and in drop tower experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The heat pipe or vapor chamber, which consists of an evaporator and a condenser, has been widely used in thermal management of electronics. However, their performance has been fundamentally limited by the physical nature of filmwise condensation (FWC). In this project, we have successfully demonstrated the effectiveness of novel Ni-Gr coatings in promoting dropwise condensation (DWC) and drastically enhancing heat pipe performance nearly an order of magnitude compared with the one with conventional heat pipes with FWC. The new heat pipe technologies can be utilized to reduce the temperature difference between the heating components such as CPUs or GPUs and coolants (air or liquid coolants), which would effectively address the bottle neck imposed by thermal management in developing next generation electronics that demand high performance. Equally important, the technology developed in this project can be extended to develop highly conductive vapor chambers that are widely used as heat spreaders for high-power devices.
Task Progress:	The dropwise condensation (DWC) enhanced heat pipes work very well with assistance of gravity. To sustain DWC inside a heat pipe, it has been extremely challenging to design novel wicking structures to return working fluid from the condensation section to evaporation section against gravity without compromising evaporating performance. It is because the wicking structure should contact the wall in the evaporation section, but a proper gap between the wicking structure and the wall should be designed in the condensation section. More challengingly, the wicking structure in both evaporation and condensation should be an integrated part in order not to interrupt liquid flow inside the wicking structure. In this report period, a novel wicking structure has been developed and sintered to create the designated gap and provide an effective liquid returning route. The wicking structure was integrated in a 42.5 cm long heat pipe. In the experiment, the wicking structure has been demonstrated to work against gravity with a peak effective thermal conductivity of 5976.6 W/m-K. However, the wicking structure in the condensation has been observed to be partially blocked. It could be caused by liquid slugs because of possible low permeability resulting from the fine copper particles. An improved design with a hybrid porous structure is being developed to address the high flow resistance in the liquid phase. If successful, it would greatly enhance the heat pipe performance in an antigravity environment.
Bibliography Type:	Description: (Last Updated: 12/21/2023)
Abstracts for Journals and Proceedings	Li C. "Improving the heat pipe performance by sustainable dropwise condensation." 7th Thermal and Fluids Engineering Conference, American Society of Thermal and Fluids Engineers (ASTFE), Las Vegas, NV, May 15-18, 2022. Abstracts. 7th Thermal and Fluids Engineering Conference, American Society of Thermal and Fluids Engineers (ASTFE), Las Vegas, NV, May 15-18, 2022. , May-2022
Articles in Peer-reviewed Journals	Chang W, Luo K, Wang P, Abdulshaheed AA, Li C. "Sustainable dropwise condensation enabled ultraefficient heat pipes." Droplet. 2023 Jan;2(1):e43. https://doi.org/10.1002/dro2.43 , Jan-2023